

# Yufan Du

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## Education

**Peking University**

Bachelor of Science in Applied Physics *and* Computer Science  
Dual Major

**Sep 2021 - Present**

*Beijing, China*

## Technical Skills

**Proficient in:** C/C++, Python

**Knowledgeable with:** Python ML libraries (PyTorch, NumPy), Linux systems, CUDA/parallel computing, Embedded systems, SQL/Database, JavaScript/HTML, Verilog/VHDL, tcl, L<sup>A</sup>T<sub>E</sub>X

**EDA tools:** Familiar with Cadence Innovus, Cadence Virtuoso.

**Tools Used:** Git, Vim, VSCode, CMake/Makefile, Linux utilities

## Relevant Scores

**Overall GPA:** 3.852 (Top 5%)

**TOEFL:** 104

**GRE:** 324 (Quantitative + Verbal), 4.0 (Analytical Writing)

## Project Experiences

### FPGA-Based Snake Game Design Project

- Developed a real-time control mechanism for the Snake game using camera-based hand movement detection.

- Enhanced player interaction by innovative human-computer interface and provided insights for IoT edge device development.

### AI-Assisted Schedule Manager

- Created an AI-integrated scheduling application with cross-platform functionality.

- Utilized natural language processing for efficient schedule management and reminders.

### Cross-Stage Power Prediction for Integrated Circuits

- Proposed an innovative framework for power prediction, incorporating cross-stage circuit-transformation-aware learning.

- Significantly reduced error rates within shorter time compared to leading commercial tools, as demonstrated on CircuitNet benchmarks.

## Relevant Publications

### Under Review:

- “PowPrediCT: Cross-Stage Power Prediction with Circuit-Transformation-Aware Learning,” submitted to *Design Automation Conference (DAC24)*.

## Relevant Coursework

### Academic Year 22-23, Semester 3

| Course Name                                    | Credits | Grade | GPA  |
|------------------------------------------------|---------|-------|------|
| Introduction to Quantum Information Technology | 2       | 96    | 3.97 |
| Aural Culture and World Civilization           | 2       | 93    | 3.91 |

**Academic Year 22-23, Semester 2**

| Course Name                                   | Credits | Grade | GPA  |
|-----------------------------------------------|---------|-------|------|
| Introduction to Computer Vision               | 3       | 95.2  | 3.96 |
| Anglo-American Tort Law                       | 2       | 90    | 3.81 |
| Computer Aided Design of Integrated Circuits  | 3       | 95.5  | 3.96 |
| Practice of Programming in C&C++              | 3       | 91    | 3.85 |
| Introduction to Database Systems              | 3       | 92    | 3.88 |
| Basic Electronics Lab                         | 2       | 92    | 3.88 |
| Advanced Design of Digital Circuit and System | 4       | 89    | 3.77 |
| Modern Physics                                | 3       | 96    | 3.97 |
| Theoretical Mechanics (B)                     | 3       | 92    | 3.88 |
| Integrated Circuit Devices                    | 4       | 86.5  | 3.66 |

**Academic Year 22-23, Semester 1**

| Course Name                                 | Credits | Grade | GPA  |
|---------------------------------------------|---------|-------|------|
| Optics                                      | 3       | 94    | 3.93 |
| Introduction to Computer Systems            | 5       | 97    | 3.98 |
| Integrated Circuit Manufacturing Technology | 3       | 92    | 3.88 |
| Analysis and Design of Analog Circuits      | 5       | 98    | 3.99 |
| Methods of Mathematical Physics             | 3       | 96    | 3.97 |
| Fascinating Chemistry                       | 2       | 91.5  | 3.87 |
| Music and Mathematics                       | 2       | 91    | 3.85 |

**Academic Year 21-22, Semester 3**

| Course Name                           | Credits | Grade | GPA  |
|---------------------------------------|---------|-------|------|
| Introduction to Cultural Anthropology | 2       | 93    | 3.91 |

**Academic Year 21-22, Semester 2**

| Course Name                                                 | Credits | Grade | GPA  |
|-------------------------------------------------------------|---------|-------|------|
| Introduction to Seismology                                  | 2       | 97    | 3.98 |
| Advanced Mathematics A (no. 2)                              | 5       | 90    | 3.81 |
| Thermal Physics                                             | 3       | 86    | 3.63 |
| Electromagnetism                                            | 4       | 99    | 4    |
| Data Structure and Algorithm (B)                            | 3       | 95    | 3.95 |
| Practical English Skills: From Listening to Public Speaking | 2       | 90    | 3.81 |

**Academic Year 21-22, Semester 1**

| Course Name                                             | Credits | Grade | GPA  |
|---------------------------------------------------------|---------|-------|------|
| Fitness                                                 | 1       | 92    | 3.88 |
| Principles of Economics                                 | 4       | 89    | 3.77 |
| Advanced Mathematics A (no. 1)                          | 5       | 84    | 3.52 |
| Linear Algebra (B)                                      | 4       | 83    | 3.46 |
| Mechanics                                               | 3       | 95    | 3.95 |
| Introduction to Computation (A)                         | 3       | 96    | 3.97 |
| An Introduction to Ideological & Moral Culture and Laws | 3       | 91    | 3.85 |
| Military Theory                                         | 2       | 94    | 3.93 |